Robust Federated Learning for Intrusion Detection Under Model Poisoning Attacks: An Exhaustive Analysis and Novel Defense Proposals

The paradigm of Intrusion Detection Systems (IDS) is undergoing a fundamental architectural shift, migrating from centralized, data-hungry models toward decentralized, privacy-preserving frameworks. Federated Learning (FL) has emerged as the premier mechanism to facilitate this transition, enabling distributed edge devices and local networks to collaboratively train deep learning models without exposing sensitive, raw packet capture (PCAP) or flow-level telemetry to a central aggregator. This decentralized approach mitigates severe privacy and regulatory compliance risks while theoretically allowing the global model to generalize across heterogeneous network topologies. However, the exact architectural decoupling that provides these privacy guarantees simultaneously introduces a catastrophic vulnerability: the central server's absolute blindness to the local training data and the local optimization process.

This foundational blindness renders Federated Learning intrinsically susceptible to adversarial manipulation, most notably through model poisoning and backdoor attacks. In these adversarial threat models, compromised clients—or maliciously injected nodes acting as Sybils—intentionally pervert their local parameter updates before transmitting them to the central server. The objective ranges from untargeted denial-of-service, where the global model's accuracy is universally degraded, to highly stealthy targeted backdoor attacks, where the global model is explicitly trained to misclassify specific malicious signatures (such as a distinct Distributed Denial of Service (DDoS) vector or a localized botnet payload) as benign traffic while maintaining normal performance on all other tasks.

Designing a robust, lightweight defense mechanism that operates efficiently within the strict computational and bandwidth constraints of Internet of Things (IoT) environments remains a critical open challenge. The following comprehensive research report critically evaluates the state-of-the-art in robust federated aggregation, identifies profound systemic gaps in current methodologies, and proposes mathematically sound, highly practical novel defense architectures explicitly tailored for tabular network data environments utilizing the PyTorch and Flower ecosystems.

1. Literature Review

The intersection of Federated Learning, cybersecurity, and Byzantine-robust aggregation has experienced explosive growth over the past six years. The literature demonstrates a continuous evolutionary arms race between increasingly sophisticated adversarial gradient manipulation techniques and highly complex server-side defense mechanisms. Table 1 synthesizes fifteen critical, high-quality research contributions that define the contemporary boundaries of this domain.

Title	Year	Core Idea	Method Used	Dataset(s)	Key Results	Limitations
Investigatin	2025	Assesses the	Systematic	UNSW-NB15	Attack impact	Focuses

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g the Impact of Label-flipping Attacks against Federated Learning for Collaborative Intrusion Detection		acute vulnerability of Federated IDS to label-flipping attacks under varying architectural hyperparameters.	quantitative assessment of targeted label-flipping perturbations within FL frameworks.	, KDD CUP 99, DÏoT	ranges from severe disruption to highly mitigated depending strictly on data distribution; mitigation remains non-trivial in highly heterogeneous settings.	predominantly on basic label-flipping heuristics rather than mathematically optimized gradient manipulation or covert backdoor strategies.
FedPrIDS: Privacy-Preserving Federated Learning for Collaborative Network Intrusion Detection in IoT	2026	Enables cross-organizational sharing of NIDS intelligence utilizing adaptive differential privacy and hybrid feature selection algorithms.	Weighted aggregation (70% data volume, 30% local loss performance) paired with Median Absolute Deviation (MAD) outlier detection.	CIC-IDS-2017, UNSW-NB15, BoT-IoT	Achieved 90.3% (CIC-IDS), 89.7% (UNSW), and 92.1% (BoT-IoT) global accuracy; demonstrated a 50% reduction in inter-node communication overhead.	Evaluated primarily on five simulated, macroscopic organizational silos; unproven scalability and latency constraints regarding thousands of resource-limited IoT edge devices.
FLARE: Defending Federated Learning Against Model Poisoning Attacks via Latent Space Representations	2025	Shifts the defensive focus entirely from raw model parameters to latent space representations to characterize adversarial influence on local updates.	Computes the Maximum Mean Discrepancy (MMD) of Penultimate Layer Representations (PLRs) utilizing an auxiliary trust dataset.	Various standardized FL benchmark sets	Defends highly effectively against semantic and trojan backdoor attacks while perfectly maintaining global model accuracy on benign data.	Relies strictly on the mathematical assumption of an honest client majority and mandates the existence of an auxiliary, clean dataset on the central server.

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FedCC: Robust Federated Learning against Model Poisoning Attacks	2023	Mitigates malicious influence in strictly non-IID settings by clustering local models based on representation similarity rather than Euclidean raw gradients.	Centered Kernel Alignment (CKA) similarity of PLRs combined with sophisticated layer-wise weighted aggregation.	Standard FL image and network benchmarks	Minimizes average degradation of global performance by 65.5%; successfully reduces targeted backdoor attack confidence to absolute zero.	Layer-wise CKA matrix computation incurs massive computational complexity and latency overhead on the central server during each communication round.
Secure Federated Learning against Model Poisoning Attacks via Client Filtering (CosDefense)	2023	Leverages the extreme sensitivity of the neural network's final classification layer to input data distributions to detect adversarial optimization momentum.	Server-side calculation of cosine similarity of the last layer's weights between global and local models; min-max normalization filtering.	MNIST, Fashion-MNIST, CIFAR-10 (Non-IID partitions)	Maintains convergence stability and faster recovery times even when the active malicious client ratio reaches the catastrophic threshold of 30-40%.	The core assumption that adversarial optimization distinctly diverges solely in the final classification layer may fail entirely against subtle, multi-layer feature-space stealth attacks.
FLTG: Byzantine-Robust Federated Learning via Angle-Based Defense and Non-IID-Aware Weighting	2025	Integrates angle-based gradient defense mechanisms and dynamic reference selection to sustain robustness under extreme statistical	ReLU-clipped cosine similarity screening utilizing a server-side clean dataset, combined continuously with inverse-proportional	MNIST, CIFAR-10	Sustains robustness and significantly outpaces state-of-the-art heuristic defenses even when over 50% of participating clients are actively	The strict dependency on a clean, highly representative server-side dataset violates the foundational zero-trust, data-isolation principles of pure

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		bias and high attacker ratios.	angular weighting.		malicious.	federated environments.
FedLAD: A Linear Algebra Based Data Poisoning Defence for Federated Learning	2025	Models the federated aggregation process strictly as a linear algebra optimization problem to filter out models that do not fit into a calculated independent basis structure.	Sub-matrix splitting and parallel optimization logic to extract independent linear combinations of submitted local models.	CIFAR10, CIFAR100, AG_NEWS	Tolerates unprecedented high ratios of malicious nodes (up to 70% in specific natural language processing tasks); mathematically outperforms Krum and Trimmed Mean.	Attack Success Rate (ASR) increases sharply and non-linearly when benign local models become too few to linearly cancel out the dominant adversarial influence.
PenTiDef: Enhancing Privacy and Robustness in Decentralized Federated Intrusion Detection Systems	2026	Eliminates central server single-point-of-failure vulnerabilities by proposing a blockchain-coordinated, fully decentralized FL-IDS architecture with integrated differential privacy.	Distributed Differential Privacy (DDP) combined dynamically with Latent Space Representations (LSR) and smart contract-based automated trust execution.	CIC-IDS2018, Edge-IIoTSet	Measurably outperforms FLARE and FedCC in advanced attack scenarios; maintains high detection accuracy without relying on any central aggregator.	Blockchain coordination naturally introduces severe latency spikes and consensus bottlenecks that render it unsuitable for highly dynamic, real-time IoT networks.
Federated learning framework for IoT intrusion detection using tab transformer	2025	Optimizes IDS data processing and embeddings specifically for tabular network	Tab Transformer (TTF) deep learning model coupled tightly with Electric Eel	N-BaIoT, UNSW-NB15, CICIoT2023	Achieved 98.02% global accuracy on the UNSW-NB15 dataset and 99.02% on	Features a prohibitively high computational and memory cost for training the TTF

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		traffic rather than relying on image-based convolutions.	Foraging Optimization (EEFO) for decentralized hyperparameter tuning.		N-BaIoT, drastically surpassing classical FL-IDS deep neural network models.	attention mechanisms locally on resource-constrained IoT edge microcontrollers.
DISFIDA: Distributed Self-Supervised Federated Intrusion Detection Algorithm	2024	Integrates self-supervised online learning continuously into the federated pipeline for adaptive anomaly detection in health IoT and high-speed vehicular networks.	Online federated updating of neural weights executed asynchronously; relies on local self-supervised traffic classification heuristics.	Health IoT telemetry, Internet of Vehicles (IoV) traffic	Maintains high detection effectiveness in highly dynamic and privacy-sensitive environments without ever transmitting raw operational logs.	Asynchronous aggregation protocols often lead to severe stale gradient issues and rapid global model drift in highly non-IID conditions.
Fed-IIoT: A Robust Federated Malware Detection Architecture in Industrial IoT	2020	An early, foundational architecture introducing robust aggregation specifically designed for Android and IoT malware detection in critical industrial settings.	Generative Adversarial Networks (GANs) utilized simultaneously for dynamic poisoning simulation (FedGAN) and countermeasure training (A3GAN).	IoT Malware datasets, CICAndMal2020	Achieves an 8% higher F1-score and accuracy metric compared to contemporary local adversarial training baselines deployed at the edge.	GAN-based adversarial training is mathematically and computationally prohibitive for standard IoT edge nodes lacking dedicated tensor processing hardware.
FL-ISM: Federated Learning-based IoT Security Model for Privacy	2026	Addresses gradient inversion, membership inference, and poisoning	Secure cryptographic aggregation, calibrated differential privacy thresholds,	Standard Intrusion and anomaly detection flow benchmarks	Successfully mitigates complex backdoor attacks while significantly reducing the	Reputation-aware selection protocols risk the severe penalty of entirely

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Preserving Analytics		attacks simultaneously while mathematically guaranteeing reduced communication overhead.	and Byzantine-resilient optimization fused with reputation-aware client selection.		uplink traffic energy overheads required by the IoT devices.	excluding minority devices that naturally generate anomalous but fully benign edge-case traffic data.
Poisoning Attacks on Federated Learning-based IoT Intrusion Detection Systems	2020	Conclusively demonstrates that FL-IDS architectures are highly vulnerable to targeted backdoor attacks initiated from seemingly low-power compromised edge devices.	Implants highly targeted backdoors by injecting microscopic amounts of malicious data via compromised clients during the local training phase.	Three distinct real-world IoT datasets collected from 46 physical devices	Successfully bypasses standard distance-based aggregation defenses, forcing the global model to confidently misclassify malicious command-and-control traffic as benign.	Focuses heavily on the mathematical proof of the attack vector rather than proposing a computationally lightweight, easily deployable counter-defense.
DyMUSA: Dynamic user selection and adaptive poisoning strategies	2025	Introduces a novel adaptive model poisoning attack that mathematically exploits systemic vulnerabilities via dynamic, shifting adversarial behavior.	Leverages Isolation Forest algorithms locally to identify anomalous client clusters and craft stealthy poisoned gradients that perfectly mimic benign variance.	Real-world recommendation and network benchmark datasets	Significantly increases target exposure (Attack Success Rate) with almost zero measurable impact on overall system accuracy, completely avoiding detection by statistical monitors.	Designed primarily for federated recommender systems, though the core adaptive poisoning mathematics translates directly to FL-IDS gradient manipulation.
PoisonShield-FL-NIDS:	2025	Integrates strict local	Client-side anomaly	FL-CNN and FedACNN	Attains 93% accuracy and	Trust-aware aggregation

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A Robust Defense Against Poisoning Attacks		anomaly filtering immediately prior to global aggregation to maintain the statistical integrity of the IDS model.	filtering combined sequentially with trust-aware global aggregation mechanisms on the central server.	architectural baselines evaluated on raw network data	a 0.96 AUC while maintaining a calculated robustness index under 0.05, even with 30% of the network acting as compromised clients.	hyperparameters must be manually and meticulously tuned per network deployment, leading to potential catastrophic drops in benign accuracy if misconfigured.

An exhaustive analysis of the literature reveals a profound evolutionary trajectory in defending FL-IDS architectures against adversarial threats. Initial defensive mechanisms relied predominantly on elementary statistical outlier rejection algorithms. Methods such as Krum, Multi-Krum, and Trimmed Mean operated on the foundational assumption that malicious gradients would manifest as Euclidean outliers positioned far from the mathematical center of mass of the benign updates. However, the literature indicates that sophisticated adversaries rapidly adapted to these defenses. By utilizing techniques such as Inner Product Manipulation (IPM) and adaptive gradient scaling, attackers developed "stealthy" model poisoning and targeted backdoors. In these advanced paradigms, the malicious gradient vector is mathematically engineered to be artificially constrained, perfectly matching the L_2 norm and expected variance of benign updates, thereby rendering distance-based statistical filters entirely obsolete.

Consequently, the state-of-the-art has forcefully shifted away from evaluating raw parameter matrices toward the analysis of Latent Space Representations (LSR) and Penultimate Layer Representations (PLR). Landmark research, notably FedCC and FLARE , established that while an adversary can mathematically disguise the final output weights of a model, the deep inner-layer representations inherently carry the unavoidable mathematical signature of the adversarial optimization path. When a model is forced to learn a backdoor trigger, the internal geometric boundaries of its feature extraction layers must warp to accommodate the anomalous rule. By analyzing these internal representations through metrics like Centered Kernel Alignment (CKA) or Maximum Mean Discrepancy (MMD), central servers can successfully detect the latent topological distortions caused by poisoning, even when the final gradient vector appears benign.

Furthermore, structural and linear algebraic defenses have recently gained significant traction. Approaches such as FedLAD treat the aggregation process not as a statistical averaging problem, but as a system of linear equations. By extracting an independent linear basis from the submitted models, these defenses can filter out malicious updates that attempt to inject orthogonal, non-conforming mathematical structures. Parallel to this, Angular and Cosine-Similarity defenses like CosDefense and FLTG have demonstrated that analyzing the angular trajectory of optimization—specifically how local updates align with the historical momentum of the global model—provides a vastly superior defense against untargeted

denial-of-service attacks compared to Euclidean distance metrics. Across all these modern methodologies, the persistent, overarching challenge remains strictly operational: balancing the immense computational complexity required to perform these mathematical verifications against the severe processing, memory, and bandwidth constraints inherent to IoT and distributed edge environments.

2. Current State of the Art

Evaluating the current state-of-the-art in FL-IDS requires a granular analysis of how modern aggregation methods intercept adversarial gradients, particularly within the context of decentralized, tabular network traffic classification. The landscape has moved far beyond rudimentary averaging, embracing complex geometrical and algebraic validations to ensure model integrity.

The best-performing defense techniques currently deployed in high-security federated environments categorize into three highly distinct mathematical methodologies. First, Latent-Space and Representation-Based Defenses have proven exceptionally resilient against stealthy, targeted backdoors. Techniques such as FLARE and FedCC extract the intermediate activation representations, typically the outputs of the dense layer immediately preceding the final softmax or sigmoid classification head. By calculating the MMD or CKA similarity of these representations across clients, the central aggregator can identify local updates that are forcing the global model to learn malicious, out-of-distribution boundary definitions. These methods are highly effective because backdoor triggers—such as a highly specific sequence of TCP flags or an anomalous packet length designed to evade detection—inevitably distort the feature extraction geometry in the latent space. Even if the attacker actively restricts the magnitude of their final gradient vector to evade Trimmed Mean or norm-clipping algorithms, the topological warping in the penultimate layers remains detectable.

Second, Angular and Cosine-Similarity Defenses have emerged to combat the well-documented "curse of dimensionality" that plagues Euclidean distance metrics in massive neural networks. Recognizing that Euclidean distance becomes mathematically meaningless when comparing parameter vectors with millions of dimensions, contemporary robust models utilize angle-based similarity metrics. Architectures like CosDefense and FLTG evaluate the cosine similarity between the historical optimization trajectory of the global model and the incoming local gradient vectors. Malicious updates, particularly those executing untargeted model degradation attacks (aiming to cause catastrophic forgetting of normal traffic patterns), consistently exhibit a cosine similarity score that diverges sharply from the established optimization momentum. FLTG advances this paradigm by incorporating dynamic reference selection and applying inverse-proportional angular weighting, mathematically scaling down anomalous vectors before they can significantly perturb the global weights.

Third, Linear Algebraic Sub-space Filtering represents the bleeding edge of Byzantine robustness. Defenses such as FedLAD treat the immense matrix of all concatenated client updates as a singular system of linear equations. By systematically applying sub-matrix splitting techniques and calculating the Reduced Row Echelon Form (RREF), the aggregation server mathematically extracts a basis of independent, benign models. This algebraic isolation allows the server to effectively ignore any malicious updates that lie outside this primary, verified subspace. Empirical evidence demonstrates that linear algebraic filtering allows the federated system to survive and converge even in catastrophic scenarios where malicious, compromised nodes vastly outnumber legitimate benign nodes—achieving up to 70% resilience in certain

highly distributed natural language processing and sequential data tasks.
To synthesize the operational mechanics of these state-of-the-art robust aggregation methods, the following table details their mathematical approaches and specific application constraints within FL-IDS.

Aggregation Method	Mathematical Mechanism	Resilience Profile	FL-IDS Application Constraint
Krum / Multi-Krum	Selects the single update (or k updates) that minimizes the sum of squared Euclidean distances to its n-f-2 nearest neighbors.	Resilient to basic outlier injection but highly vulnerable to dense Sybil attacks and collusion.	Discards massive amounts of valid benign information; performs poorly on highly non-IID network traffic partitions.
Trimmed Mean / Median	Sorts parameters dimension-by-dimension, discarding the extreme \beta fractions before averaging, or simply extracting the median.	Highly resilient to single-parameter spikes and basic untargeted gradient explosion.	Fails completely against stealthy attacks that keep parameter values within the expected statistical variance but alter their correlation.
Cosine Momentum Weighting	Calculates the angle between the incoming gradient vector and the historical exponential moving average of the global model.	Exceptional resilience against untargeted denial-of-service and catastrophic forgetting attacks.	May struggle with highly dynamic, rapidly shifting zero-day attack landscapes where legitimate optimization momentum must suddenly pivot.
Latent-Space MMD/CKA	Projects an auxiliary dataset through the local models, calculating the statistical discrepancy of the resulting activation maps.	Unparalleled resilience against sophisticated, norm-bounded targeted backdoors and semantic triggers.	Requires immense server-side computational overhead and theoretically violates strict zero-knowledge privacy guarantees.
Sub-space Filtering (RREF)	Extracts the independent linear basis of the concatenated update matrix, zeroing out orthogonal adversarial vectors.	Survives extreme majority-malicious scenarios (e.g., >50% compromised nodes).	Computationally intensive matrix factorization operations; scales poorly with massive deep neural network parameter counts.

Under active attack, the performance of modern FL-IDS architectures exhibits stark contrasts depending on the defense utilized. When subjected to advanced Model Poisoning Attacks (MPA), an unprotected FL-IDS utilizing standard Federated Averaging (FedAvg) collapses almost instantly. In targeted label-flipping scenarios applied to the UNSW-NB15 dataset, empirical evaluations show FedAvg accuracy plummeting from a baseline of ~96% to below 50% within a few communication rounds. However, when shielded by state-of-the-art defenses, these architectures demonstrate remarkable resilience. Systems deploying Tab Transformers

(TTF) with federated hyperparameter tuning maintain ~98-99% accuracy on N-BaIoT and UNSW-NB15 even under adversarial pressure. Furthermore, when subjected to active, targeted backdoor attacks executing Inner Product Manipulation (IPM), angular defenses like CosDefense have proven capable of recovering model utility and forcing convergence even when 30% to 40% of the network nodes are actively injecting poisoned gradients. The state-of-the-art conclusively proves that robustness is mathematically achievable, albeit currently at the cost of immense computational overhead and restrictive data distribution assumptions.

3. Gap Analysis

Despite the rapid, mathematically sophisticated advancements in federated adversarial defense, a critical and exhaustive analysis of the literature reveals severe, unaddressed gaps. These gaps are specifically pronounced when attempting to transition these theoretical defenses from academic, image-based benchmarks into real-world, resource-constrained IoT environments tasked with processing tabular network telemetry.

The most glaring vulnerability in the current state-of-the-art is the "Clean Server Data" fallacy. A prevailing operational requirement in the highest-performing defenses—such as FLARE , FLTG , and various MMD-based latent space analyzers—is the absolute necessity for an "auxiliary" or "clean" root dataset hosted permanently on the central aggregation server. This dataset is required to compute reference representations, baseline cosine similarities, or kernel alignments to evaluate the incoming local updates. In a true zero-trust, privacy-first FL-IDS deployment, the central server (frequently a multi-tenant cloud aggregator) is explicitly and legally forbidden from holding raw, unencrypted network traffic data or historical PCAPs. If the server lacks a highly representative, clean dataset containing both contemporary benign and attack traffic to evaluate these similarities, these advanced defense mechanisms are entirely neutralized and cannot execute their anomaly detection logic. This reliance on server-side data fundamentally violates the core cryptographic and privacy-preserving tenets that motivate the use of Federated Learning in the first place.

Compounding this architectural flaw is the profound issue of Non-IID Amplification of False Positives. Network traffic is notoriously non-Independent and Identically Distributed (non-IID) across physical geographical boundaries and distinct industrial sectors. An FL-IDS client deployed within a highly regulated financial sector subnet will continuously observe vastly different baseline traffic metrics and specific attack signatures (e.g., highly targeted, low-volume SQL injections or specialized application-layer attacks) compared to an IoT edge sensor deployed in an automated smart factory (which may routinely encounter high-volume Modbus protocol flooding or localized Mirai botnet scans). When robust aggregators—ranging from legacy Krum to advanced angular weighting systems—measure mathematical distance or angular divergence to filter outliers, they inadvertently and severely penalize benign clients that happen to hold minority classes or highly unique, non-IID data distributions. The defense mechanism classifies the legitimate, highly specialized local update as a "poisoned gradient" because it diverges from the global statistical norm. Consequently, the global IDS model is forced into a state of generalized mediocrity, entirely ignorant of localized, specialized attacks, thereby defeating the purpose of collaborative threat intelligence.

A third major gap lies in Dimensionality and Tabular Data Friction. An overwhelming majority of robust FL defenses present in the literature—including CosDefense, FedLAD, and FedCC—were conceptualized, mathematically proven, and evaluated exclusively on image

classification datasets such as MNIST, Fashion-MNIST, and CIFAR-10. Image data possesses dense, continuous spatial hierarchies that are highly suitable for Convolutional Neural Networks (CNNs). In stark contrast, Intrusion Detection Systems rely almost exclusively on discrete, tabular data (e.g., CIC-IDS2017 feature sets encompassing flow duration, packet length variance, and TCP flag counts). Algorithms that rely heavily on the topological warping of Penultimate Layer Representations (PLRs) often fail catastrophically to capture semantic anomalies in tabular feature spaces. Standard feedforward neural networks and multi-layer perceptrons (MLPs) do not project tabular data into latent spaces with the same geometric stability, density, or continuous variance as CNNs processing high-resolution images. Applying an image-proven defense to a tabular IDS dataset frequently results in unpredictable behavior and degraded adversarial resilience.

Finally, the literature exhibits a near-total disregard for Implementation and Scalability Latency at the edge. The mathematical complexity of advanced defenses restricts their real-world IoT deployment. For example, computing the Centered Kernel Alignment (CKA) pairwise similarity matrix for N clients, as required in FedCC, scales at a computational complexity of O(N^2) with respect to the dimensionality of the specific neural network layer being analyzed. On a central server attempting to manage and aggregate updates from 10,000 distinct IoT clients simultaneously, this quadratic scaling introduces extreme latency. This processing bottleneck completely prevents the high-frequency, real-time global model updates necessary to successfully combat and propagate defenses against rapidly evolving, zero-day network intrusions.

4. Shortcomings in Existing Work

Understanding precisely why highly motivated attackers still succeed against these sophisticated architectures requires a critical examination of the inherent mathematical vulnerabilities and systemic trade-offs accepted by current federated defense paradigms. The shortcomings in existing work stem not from a lack of mathematical rigor, but from fundamentally exploitable operational assumptions.

Current defenses fail profoundly when subjected to advanced Collusion and Sybil Attacks. Defenses such as Krum, Trimmed Mean, and even basic angular filtering are designed around the core assumption that malicious updates are singular, independent statistical outliers scattered randomly at the fringes of the parameter space. However, intelligent adversaries entirely bypass this by executing coordinated Sybil attacks. If an attacker successfully compromises a subset of FL-IDS nodes—a highly realistic scenario in the context of massive IoT botnets, as observed in successive Mirai variants—they can meticulously synchronize their poisoned gradient updates. By submitting identical, or tightly clustered, malicious gradients that share the exact same optimization vector, the attackers artificially construct a dense, mathematically coherent center of mass within the parameter space. Aggregation algorithms attempting to identify the "nearest neighbors" or the statistical median will erroneously identify this massive malicious cluster as the benign majority consensus. Consequently, the defense mechanism will happily shift the global model toward the attacker's adversarial objective while actively discarding the genuinely scattered benign updates as untrusted "outliers."

Furthermore, existing literature frequently ignores the mathematically proven Accuracy vs. Robustness Trade-off. There exists a theoretical lower bound demonstrating an unavoidable, zero-sum trade-off between standard classification accuracy on benign data and guaranteed adversarial robustness. When an FL-IDS employs aggressive defensive

countermeasures—such as severe gradient norm clipping, high-variance differential privacy noise injection, or heavy regularization specifically designed to suppress backdoor triggers—it simultaneously and unavoidably restricts the neural network's fundamental capacity to learn nuanced, highly complex patterns of sophisticated cyber attacks. For instance, defending against a massive label-flipping attack might completely blind the model to the subtle, low-rate traffic patterns characteristic of a slow-loris DDoS attack or a covert data exfiltration channel. Consequently, while the global model mathematically survives the poisoning attempt, its baseline False Negative Rate (FNR) for genuine, non-adversarial intrusions increases to levels that are practically unacceptable for enterprise-grade security deployments.

Modern attackers also succeed by exploiting Stealth via Gradient Projection, bypassing defenses that rely on magnitude or simple thresholding. Adaptive adversaries no longer blindly flip labels or inject massive, easily detectable gradient magnitudes that trigger Trimmed Mean filters. In advanced attack vectors like DyMUSA or Inner Product Manipulation (IPM) , the adversary meticulously projects their malicious gradient precisely into the null space of the benign updates. Alternatively, they actively constrain the L_2 norm of their calculated poisoned update to perfectly match the expected, historical norm of a verified benign client. Because the absolute magnitude of the vector is mathematically identical to benign traffic, norm-clipping defenses completely ignore the update. Because the update moves the model parameters along an orthogonal adversarial vector without significantly disturbing the primary task performance on the main training distribution, it easily evades validation accuracy checks. This allows the attacker to implant stealthy backdoors—such as ensuring that network traffic originating from a specific, hardcoded malicious IP address is perpetually classified as benign—without ever triggering an alert.

Lastly, Asynchronous Vulnerabilities remain largely unaddressed. Works aiming to solve the intermittent connectivity issues of IoT devices, such as DISFIDA , heavily promote asynchronous federated aggregation. However, asynchronous aggregation inherently engineers a permanent vulnerability window. Attackers can intentionally and strategically delay the submission of their poisoned updates, silently waiting for the global model to reach a state of mature convergence. Once the global weights are stabilized, the attacker submits a highly tailored, massive update specifically crafted to pivot the mature global weights. Because the server is aggregating asynchronously, this sudden, massive adversarial vector easily overpowers the smaller, incremental benign updates arriving sporadically from resource-constrained, honest edge nodes.

5. Novel Ideas

To directly address the identified gaps—specifically the failure of image-centric defenses on non-IID tabular network data, the strict privacy requirement prohibiting server-side auxiliary datasets, and the severe computational latency constraints of IoT environments—I propose three mathematically strong, highly practical novel defense architectures. These approaches are explicitly designed to be feasible for an advanced capstone implementation utilizing the PyTorch and Flower frameworks, dramatically improving robustness against state-of-the-art poisoning in FL-IDS settings.

Idea 1: Adaptive Layer-Wise Cosine Momentum Tracking (AL-CMT)

Intuition: Current robust aggregators either compare client updates against each other (which

fails catastrophically under Sybil collusion) or against a server-held clean dataset (which violates strict federated privacy laws). AL-CMT posits that the server should evaluate a client's incoming update strictly against *its own historical momentum* and the *global model's established historical trajectory*. A malicious client attempting to implant a backdoor or artificially force catastrophic forgetting must fundamentally alter the optimization direction of the network's classification layers. By tracking the momentum of the angle of optimization, rather than the Euclidean distance of the weights, we can isolate adversarial intent without needing to see the underlying data.

Architecture and Design:

1. The central Flower Server maintains an Exponential Moving Average (EMA) of the global gradient vector: $m_t = \beta m_{t-1} + (1-\beta) \Delta \theta_{global}$, where β controls the momentum decay.
2. During communication round t, the server receives the local parameter update $\Delta \theta_i$ from client i.
3. Instead of analyzing the entire massive model, the server isolates only the final two layers of the PyTorch neural network (the classification heads). These specific layers are mathematically proven to be the most sensitive to label flipping and targeted backdoors in tabular datasets.
4. The server computes the Cosine Similarity S_i between the client's isolated classification layer update and the corresponding layers in the global momentum vector m_t.
5. **Dynamic Thresholding:** Instead of utilizing a rigid, hardcoded cutoff threshold (which attackers can easily calculate and evade), the server dynamically calculates the Median Absolute Deviation (MAD) of all S_i scores received in the current round. Clients whose cosine similarity falls outside the dynamic boundary of $3 \sigma_{MAD}$ are immediately flagged as adversarial and assigned an aggregation trust weight of 0. Surviving updates are scaled proportionally to their similarity score before standard averaging.

Why it Improves Over Existing Work: AL-CMT entirely eliminates the need for a clean server dataset, solving the privacy gap present in FLARE and FLTG. By utilizing MAD on cosine similarity scores rather than raw Euclidean distances, it is highly immune to the "curse of dimensionality" and resistant to Sybil clusters. Furthermore, it inherently supports non-IID network data; benign non-IID updates, while variant in magnitude, still generally align with the global optimization direction toward generalized accuracy over time. In contrast, targeted poisoning mandates a sudden, extreme angular divergence that AL-CMT easily intercepts.

Complexity: Low. Calculating cosine similarity on a tightly restricted subset of the outermost layers is computationally inexpensive $O(d_{subset})$, entirely avoiding the massive $O(N^2)$ overhead of full-model pairwise comparisons required by methods like FedCC.

PyTorch + Flower Implementation: This requires overriding the aggregate_fit method within a custom Flower Strategy class (e.g., inheriting from flwr.server.strategy.Strategy).

- Extract the state dictionaries from the incoming results: List].
- Iterate through the dictionaries, isolating only the keys corresponding to the final nn.Linear layers.
- Convert these specific layer arrays into PyTorch tensors using torch.tensor().
- Compute torch.nn.functional.cosine_similarity against the stored global momentum tensor.
- Calculate the MAD using standard numpy.median operations, filter the results list to drop malicious proxies, and pass the weighted, sanitized remnants to the standard aggregate helper function to formulate the new global parameters.

Idea 2: Client-Side Autoencoder Reconstruction Filtering (CS-ARF)

Intuition: A profound limitation of FL-IDS is the central server's inability to determine client trust without visually inspecting the training data. If a compromised client is actively training on poisoned, synthetic network traffic designed to implant a backdoor, the underlying statistical distribution of that specific traffic batch fundamentally differs from the natural variance of benign data. CS-ARF forces clients to mathematically "prove" the normality of their local data distribution via an unsupervised auxiliary task before their gradients are accepted.

Architecture and Design:

1. Alongside the primary supervised IDS classifier, an extremely lightweight PyTorch Autoencoder (AE) is distributed to all participating clients.
2. During the local fit phase, clients must first train the AE locally to reconstruct their raw network traffic features unsupervised. They then train the supervised IDS classifier. The client returns both the IDS classifier gradients and the separate AE gradients to the server.
3. The central server securely aggregates the AE updates to create a highly generalized "Global Representation of Normality."
4. In the subsequent round, prior to training, clients must calculate their average Reconstruction Error (RE) using the newly downloaded Global AE on their local current training batch. They attach this scalar RE value as a custom metric inside their Flower FitRes return dictionary.
5. If an attacker is attempting to inject highly anomalous backdoor traffic (e.g., synthetically generated DDoS packets with impossible TCP flag combinations), the Global AE will fail to reconstruct it, yielding a significantly higher RE scalar on that specific client's dataset. The server reads these metrics and scales the client's aggregation weight for the IDS classifier inversely proportional to their reported RE.

Why it Improves Over Existing Work: This architecture brilliantly shifts the complex anomaly detection burden entirely to the client side using unsupervised learning, completely resolving the server-side data privacy dilemma. It directly addresses the tabular nature of datasets like CIC-IDS2017 and UNSW-NB15, as Autoencoders are mathematically proven to be highly effective at capturing dense tabular feature correlations, succeeding where image-centric latent space metrics fail.

Complexity: Medium. It strictly requires training two separate models (Classifier + AE) on the edge device. However, because it operates on tabular data, the AE can be extremely small and shallow (e.g., a simple architecture of 78 -> 32 -> 78 neurons for the CIC-IDS2017 feature set), resulting in negligible memory overhead.

PyTorch + Flower Implementation:

- Modify the PyTorch ClientApp training loop. The fit() function must sequentially execute the PyTorch optimizer step for the AE, calculate the Mean Squared Error (MSE) reconstruction loss, and then execute the training loop for the primary Classifier.
- Pass the calculated recon_error scalar back to the server inside the metrics dictionary argument of the FitRes return object.
- In the custom Strategy running on the Server, extract metrics["recon_error"] for each client. Apply a softmax-based inverse weighting function to determine the multiplier for each client's parameters during the aggregation step.

Idea 3: Singular Spectrum Filtering of Gradients (SSFG)

Intuition: Inspired by the linear algebra optimization of FedLAD , but heavily optimized for computational speed on central servers. If we mathematically stack all incoming client parameter updates into a massive 2D matrix, the primary principal components of that matrix represent the generalized, consensus knowledge of the entire federated network. Poisoned updates, which introduce highly anomalous, specific, and orthogonal mathematical features (such as backdoors), will inevitably map to the lower-variance, noise-level singular vectors.
Architecture and Design:

1. During aggregation, the server flattens the multidimensional parameter updates from N clients into a single 2D matrix $M \in \mathbb{R}^{N \times D}$ (where D represents the total number of flattened model parameters in a specific layer).
2. The server performs a rapid, truncated Singular Value Decomposition (SVD) on this matrix: $M \approx U \Sigma V^T$.
3. By mathematically reconstructing the update matrix utilizing only the top k singular values (which represent the dominant, consensus benign gradients of the fleet), the server effectively filters out the high-frequency noise and the orthogonal, malicious targeted perturbations that reside exclusively in the discarded lower singular values.
4. The newly aggregated global model is simply the arithmetic mean of this reconstructed, mathematically filtered matrix.

Why it Improves Over Existing Work: Unlike Krum, Trimmed Mean, or Median, which operate by outright discarding entire clients (losing all their valid data), SSFG mathematically extracts the benign mathematical components *out* of a poisoned client's update. Even if a client is 50% malicious, their useful benign gradient information regarding normal traffic is retained, while the orthogonal adversarial backdoor vector is cleanly filtered out by the SVD truncation.
Complexity: High in terms of memory utilization, but highly manageable if applied strictly layer-by-layer rather than attempting to perform SVD on the entire flattened deep neural network simultaneously.
PyTorch + Flower Implementation:

- Within the aggregate_fit method of the custom Strategy, extract the PyTorch parameter arrays and flatten them into a 2D numpy.ndarray.
- Utilize scipy.sparse.linalg.svds to perform a highly optimized truncated SVD layer-by-layer, requesting only the top k singular values (where k is a hyperparameter based on the expected benign variance).
- Reconstruct the filtered matrix using the dot product of the truncated U, Σ, and V^T matrices.
- Average the rows of the reconstructed matrix, reshape them back to the original PyTorch tensor dimensions, and formulate the new global model parameters.

6. Proposed Solution Refinement

A critical review of the provided capstone project draft indicates a fundamentally sound structural foundation. However, to elevate the project to publication-level rigor and ensure the defense mechanism succeeds in a highly volatile, non-IID, tabular data context, specific operational refinements are strictly required.

Refinement of the Defense Mechanism

The original draft vaguely proposes "detecting abnormal model updates using gradient similarity metrics" or applying "client reputation or trust scoring."

- **Crucial Improvement:** Generic Euclidean similarity metrics fail entirely against intelligent attackers executing Sybil or norm-bounded stealth attacks. It is highly recommended to abandon basic distance metrics and adopt **Idea 1 (Adaptive Layer-Wise Cosine Momentum Tracking - AL-CMT)**. By explicitly focusing on the Cosine Similarity of the final classification layer compared against the historical global momentum, the project establishes a highly specific, mathematically defensible boundary. This approach is lightweight enough to simulate IoT environments effectively while providing absolute resilience against state-of-the-art stealth attacks that spoof gradient magnitudes.

Refinement of Attack Simulations

The draft mentions broad attack categories: gradient manipulation, label flipping, and backdoor attacks. To demonstrate true academic rigor, these must be simulated using advanced, targeted methodologies.

- **Crucial Improvement:**
    1. **Label Flipping:** Avoid randomized label flipping, as standard FedAvg naturally averages out random noise. Instead, simulate a *Targeted Semantic Label Flip*. For example, explicitly code the malicious clients to flip all instances of "DDoS" traffic to "Benign," while leaving all other attack classes (like "PortScan" or "Infiltration") completely untouched. This specifically tests if the defense can catch subtle, highly focused model degradation.
    2. **Backdoor Attack:** In tabular network data, a backdoor trigger is not a visual pattern of pixels. Simulate a backdoor by forcing a highly specific, anomalous feature value combination (e.g., hardcoding Flow Duration to exactly 999999 and ACK Flag Count to 1) and systematically mislabeling these specific injected samples as benign traffic. The malicious client must then scale their outgoing gradient norm to perfectly match the benign network average, simulating a *stealth* attacker bypassing basic distance anomaly detection.

Refinement of Evaluation Metrics

Evaluating an IDS based solely on "accuracy" is fundamentally deceptive, as normal background traffic heavily outweighs attack traffic, leading to the accuracy paradox.

- **Crucial Improvement:** The experiment must track and plot the following advanced metrics:
    1. **Macro F1-Score:** To ensure the minority, highly dangerous attack classes are successfully learned and retained by the global model.
    2. **Attack Success Rate (ASR):** Essential for evaluating the backdoor attack. This metric measures exactly what percentage of the injected tabular triggers are successfully and erroneously misclassified by the global model as benign. A robust defense must force the ASR to near 0%.
    3. **False Positive Rate (FPR):** A defense is practically useless in a real-world Security Operations Center (SOC) if it causes the model to flag all normal IoT traffic as

malicious, creating alert fatigue.

Architecture Diagram Description

The final report should feature a detailed architectural diagram, conceptually structured as follows:

- **Tier 1: IoT Client Node (PyTorch Edge):** The physical edge device ingests raw network PCAP data. The data flows through a Feature Extraction and Preprocessing module (applying Z-score Standardization and Min-Max Scaling). The tabular data is fed into a PyTorch Multi-Layer Perceptron (MLP). The local PyTorch training loop executes, calculating the local gradients based on the local loss function.
- **Tier 2: FL Communication Layer (Flower):** The client application wraps the PyTorch model within the flwr.client.NumPyClient class. It serializes the weight parameters into byte streams and transmits them securely via gRPC protocols over the network to the Central Server.
- **Tier 3: Central Server Node (Robust Aggregation):** The Flower Server receives the FitRes objects from the fleet. The custom PyTorch Strategy class intercepts the weights, invoking the aggregate_fit method. It extracts the final layer tensors, calculates the Cosine Momentum against the historical cache, filters out extreme statistical outliers using the Median Absolute Deviation (MAD), applies weighted averaging to the surviving tensors, updates the Global PyTorch Model, evaluates global metrics, and broadcasts the sanitized weights for the next round.

7. Experiment Plan

To conclusively prove the mathematical efficacy of the proposed defense mechanism, the experimental setup must be exceptionally rigorous, fully reproducible, and accurately reflect realistic, constrained network topologies.

Datasets and Preprocessing

- **Datasets:** The experiments will utilize **CIC-IDS2017** (representing general enterprise and cloud network traffic) and **BoT-IoT** (representing highly specific, constrained IoT network threats).
- **Feature Selection:** Raw network data contains immense mathematical redundancy. Prior to federated training, apply rigorous Variance Filtering and Pearson Correlation analysis to reduce the CIC-IDS2017 dataset from its original 78 features to a highly discriminative subset of ~40-50 features. This simulates the lightweight processing capabilities of actual IoT edge microcontrollers. All features must be normalized using strict Z-score standardization to ensure gradient stability.

Data Partitioning (IID vs. Non-IID Setup)

The evaluation must strictly test both Independent and Identically Distributed (IID) and non-IID conditions to validate real-world applicability.

- **IID Partitioning:** The standardized dataset is shuffled globally and distributed perfectly evenly across all simulated clients, ensuring every client has an identical statistical

representation of all attack classes.

- **Non-IID Partitioning (Crucial):** Real networks are profoundly heterogeneous. Use a **Dirichlet Distribution** (\alpha = 0.5) over the attack labels to partition the data. This mathematical partitioning forces some clients to observe almost entirely DoS attacks, while others observe mostly Botnet or benign traffic. This simulates real-world, isolated physical subnetworks and rigorously tests if the robust defense accidentally drops valid edge-case data.

Network Topology & Malicious Ratios

- **Scale:** Simulate a total federated network of N = 50 distinct clients. During each communication round, randomly sample a fraction of C = 20 active clients for training.
- **Threat Vectors:** Execute distinct experimental runs representing varying severity of compromise: 0% malicious clients (Baseline performance), 10% (Low, covert threat), 30% (High, coordinated threat), and 50% (Extreme Sybil attack scenario).

Baselines for Empirical Comparison

To prove the superiority of the novel mechanism (AL-CMT or SSFG), it must be plotted directly against existing industry standards on the same axes:

1. **Standard FedAvg:** Demonstrates the baseline, unprotected vulnerability of the system, confirming the attacks are functional.
2. **Fed-Trimmed Mean:** Operates by trimming the top and bottom 20% of parameter values. Highlights the failure of basic magnitude-based filtering.
3. **Krum / Multi-Krum:** Represents the standard Byzantine-robust, Euclidean distance-based metric. Highlights the failure of distance metrics in high-dimensional non-IID spaces.

Structured Execution Flow

The experiment will proceed in three distinct, measurable phases:

1. **Phase 1 (Convergence Baseline):** Train the global model for the first 10 communication rounds with zero attackers active. This establishes the absolute baseline Macro F1-Score and proves the PyTorch model architecture is sound.
2. **Phase 2 (Adversarial Injection):** From round 11 to 30, activate the designated malicious clients. They will systematically submit stealthy backdoor updates and targeted label flips perfectly engineered to evade basic detection.
3. **Phase 3 (Evaluation and Plotting):** Continuously measure and plot the drop in the Global F1-Score and the simultaneous spike in the Attack Success Rate (ASR) across all tested aggregation methods. A mathematically successful capstone execution will visually demonstrate that under the proposed novel defense, the ASR remains suppressed near 0%, and the global F1-score maintains stability, completely avoiding the catastrophic degradation observed in standard FedAvg and Krum deployments.

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